

Matching As An Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Programme

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This paper considers whether it is possible to devise a nonexperimental procedure for evaluating a prototypical job training programme. Using rich nonexperimental data, we examine the performance of a two-stage evaluation methodology that (a) estimates the probability that a person participates in a programme and (b) uses the estimated probability in extensions of the classical method of matching. We decompose the conventional measure of programme evaluation bias into several components and find that bias due to selection on unobservables, commonly called selection bias in econometrics, is empirically less important than other components, although it is still a sizeable fraction of the estimated programme impact. Matching methods applied to comparison groups located in the same labour markets as participants and administered the same questionnaire eliminate much of the bias as conventionally measured, but the remaining bias is a considerable fraction of experimentally-determined programme impact estimates. We test and reject the identifying assumptions that justify the classical method of matching. We present a nonparametric conditional difference-in-differences extension of the method of matching that is consistent with the classical index-sufficient sample selection model and is not rejected by our tests of identifying assumptions. This estimator is effective in eliminating bias, especially when it is due to temporally-invariant omitted variables.

1. INTRODUCTION

This paper addresses the following question. It is *possible to devise a nonexperimental procedure for evaluating a prototypical job training programme that produces impact estimates and inferences about the programme that are very close to those produced from a randomized social experiment?* We combine nonexperimental data on persons who chose not to participate in the programme with data from a large scale social experiment to examine the performance of various matching methods, including new conditional difference-in-differences extensions of matching methods that we present here, in estimating an averaged version of the effect of treatment on the treated.

Matching methods pair programme participants with members of a nonexperimental control group who have similar observed attributes and estimate treatment impacts by subtracting mean outcomes of matched comparison group members from the mean outcomes of matched participants. We extend traditional matching methods by (a) incorporating exclusion restrictions across programme outcome and programme participation equations; (b) presenting weaker conditions under which matching is justified as an evaluation method than appear in the published literature; (c) incorporating prior information about the functional form of estimating equations, including additive separability between measured and unmeasured determinants of outcomes; (d) extending matching to a longitudinal context and producing a generalized difference-in-differences estimator that identifies parameters of interest under different, and generally weaker, assumptions than matching, and (e) providing a rigorous asymptotic distribution theory for the matching estimator under general conditions without invoking special assumptions about the distribution of the data.

Using data from the control group of an experiment in conjunction with data from several nonexperimental comparison groups, this paper tests the assumptions that justify our approach and the stronger assumptions that are traditionally invoked to justify matching methods. We reject the strong assumptions maintained in the literature but find support for the weaker assumptions that justify our generalized difference-in-differences extension of matching. Matching methods reduce the conventional measure of bias substantially for most groups but do not eliminate it entirely. Our generalized difference-in-differences estimator is generally more effective than conventional matching methods in removing bias from our data, especially when it is contaminated by temporally-invariant components of bias such as unobserved site and questionnaire effects.

We also address a second question. *What features of the nonexperimental data and of the matching method are essential in reducing the conventional measure of bias used in evaluation studies?* To address this question, we take as our point of departure the observation that ideal social experiments identify programme impacts by balancing many features of the data at the same time: (1) Participants and controls have the same distributions of unobserved attributes; (2) They have the same distributions of observed attributes; (3) The same questionnaire is administered to both groups, so outcomes and characteristics are measured in the same way for both groups; and (4) Participants and controls are placed in a common economic environment.

Features (2)–(4) can also be achieved in a nonexperimental evaluation. Resampling methods can be applied to nonexperimental data to mimic feature (2) of an experiment. Matching methods substantially reduce bias when features (3) and (4) characterize the nonexperimental data. When they do not, the conventional matching method can fail dramatically, as we demonstrate. However, our difference-in-differences extension of matching is more robust than conventional methods in data in which features (3) and (4) are absent.

The recent econometric literature on programme evaluation has emphasized feature (1)—elimination of selective differences in unobservables drawn from a common distribution for participants and experimental controls—as the principal benefit of randomized trials.¹ Our study suggests that this emphasis is misplaced. Features (2)–(4) are far more important to the success of the experimental method in evaluating the training programme we study than is feature (1). Selection bias, rigorously defined, is a *relatively* small part of bias as conventionally measured.

1. See, e.g. Ashenfelter (1978), Ashenfelter and Card (1985), and LaLonde (1986).

This paper emphasizes the interplay between data and method. Both matter in evaluating the impact of training on earnings for four different demographic groups. The consistency of our findings across these groups is striking. The effectiveness of any econometric estimator is limited by the quality of the data to which it is applied, and no programme evaluation method “works” in all settings. We produce some striking examples where estimators that perform well on good data perform poorly when applied to bad data. Failure to locate participants and comparison group members in the same labour market is a major source of evaluation bias; so is the failure to use the same definitions of outcome and explanatory variables, as often occurs when different surveys are administered to participants and comparison group members. Estimation methods also matter. Simple balancing of observables in the participant and comparison group samples goes a long way toward producing a more effective evaluation strategy.

This paper develops in the following way. Section 2 defines the evaluation problem and discusses the benefits of randomized experiments compared to nonexperimental methods. Section 3 shows how matching approximates randomization and presents the identifying assumptions that justify the method. Section 4 presents several extensions of the classical method of matching and the identifying assumptions that justify them. Section 5 summarizes the main findings from our previous empirical research. In that research, we use nonparametric methods to characterize the form of the evaluation bias that arises from using members of a comparison group of self-selected nonparticipants to proxy what participants would have experienced if they had not participated in the programme we study. We discuss the implications of this research for the design of a successful evaluation strategy. This evidence motivates our extensions of the method of matching. Section 6 reviews our evidence on the determinants of programme participation. Section 7 defines the main samples and models used in this paper. Section 8 demonstrates that the distributions of the matching variables are different for participant and comparison group samples. This finding has significant implications for understanding the sources of evaluation bias as conventionally measured. Section 9 decomposes conventional measures of evaluation bias into components due to (a) selection on unobservables, (b) failure to compare participants and controls at common values of matching variables, and (c) failure to weight the two groups comparably. We show that bias due to mismatching and misweighting of the data is numerically more important than bias due to selection on unobservables. Comparing the incomparable is a major source of evaluation bias. Yet the bias left over after adjusting for weighting and mismatching is still large compared to the experimentally-estimated treatment impact. Matching reduces bias but is not guaranteed to produce reliable estimates of programme impacts. Our difference-in-differences extension of the method of matching is usually more effective than conventional methods in the samples we analyse. Section 10 uses the experimental data to test the identifying assumptions invoked in the matching literature and our various extensions of it.

Section 11 defines a class of matching estimators and longitudinal and repeated cross section difference-in-differences estimators within a unified framework and Section 12 compares the empirical performance of alternative matching methods and our extensions of them. We measure the effectiveness of a nonexperimental estimator by how well it eliminates differences in earnings between a nonexperimental comparison group and a randomized-out control group.

We examine the features of the data that attenuate bias in our samples. Section 13 studies the effectiveness of different matching estimators when the probabilities of participation on which the matches are based are estimated with progressively coarser conditioning information. Several estimators perform moderately well for all demographic

groups when data on recent labour market histories are included in estimating the probability of participation, but not when earnings histories or labour force histories are absent.

Section 14 analyses comparison group samples drawn from SIPP data to assess the importance of controlling for geographical location and of using the same survey instrument to collect comparison group data. Our results indicate that geographical proximity and uniformity of the survey instrument across treatment and comparison group samples are necessary features of a successful evaluation study of earnings impacts. This evidence confirms the importance of local labour markets in determining wages—a point emphasized in the research of Blanchflower and Oswald (1994). The major source of bias arising from the application of nonexperimental estimators to evaluate training programmes that is reported in LaLonde (1986) arises from the mismatch of questionnaires and labour markets across treatment group and comparison group members, and not because of the failure of econometric estimators to eliminate selection bias.

Section 15 investigates the performance of matching estimators when dropouts or “no-shows” are used as a comparison group. Programme dropouts are located in the same labour market and are administered the same questionnaire as programme completers, so this group automatically satisfies two key requirements for a successful nonexperimental evaluation. Our evidence on the performance of “no shows” as a comparison group is mixed. Section 16 summarizes the evidence.

2. THE EVALUATION PROBLEM AND THE BENEFITS OF RANDOMIZED EXPERIMENTS

The evaluation problem is a missing data problem. At any time, persons may be in either one of two potential states but not in both.² The states associated with receiving treatment and not receiving treatment are denoted “1” and “0” respectively. Outcomes are (Y_1, Y_0) . Let $D=1$ if a person is in state “1”; $D=0$ otherwise. The outcome observed for an individual is Y defined as

$$Y = DY_1 + (1 - D)Y_0.$$

This is the Fisher model (1951), the Roy Model (1951) or the switching regression model of Quandt (1972).³ The gain from participating in the programme is $\Delta = Y_1 - Y_0$. If we could simultaneously observe Y_1 and Y_0 for the same person, there would be no evaluation problem since one could construct Δ for everyone.

To cast the discussion in familiar econometric notation, write outcomes as a function of observables (X) and unobservables (U_1, U_0)

$$Y_1 = g_1(X) + U_1, \quad (1a)$$

$$Y_0 = g_0(X) + U_0. \quad (1b)$$

The conventional econometric approach maintains that $E(U_1|X) = 0$ and $E(U_0|X) = 0$ and further assumes that g_1 and g_0 are nonstochastic functions. For the familiar case of linear regression, the g functions specialize to $g_1(X) = X\beta_1$, and $g_0(X) = X\beta_0$. These functional form assumptions are not required to implement matching estimators, but we use them in certain semiparametric extensions of the method of matching.

2. For simplicity we consider the two outcome case. Extension to a multiple-outcome switching model is straightforward.

3. Statisticians sometimes call this the “Rubin model” after a clear exposition of Fisher’s model of experiments presented by Rubin (1978).